Reprogramming Across the Organ Boundary

A graded rho model of OSK partial reprogramming in hindbrain motor neurons: dynamic weight maps for epigenetic state, the induction schedule as a scoped σ-intervention, and a proofreading bound on achievable selectivity

L. G. Meredith
F1R3FLY.io

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Scope. This note is a preclinical modelling analysis. It specifies a computational model, the fragment of the calculus that hosts it, the measurements the model requires, and the claims the model is and is not competent to support. It is not a protocol, a dose, a delivery specification, or a clinical recommendation, and nothing in it should be read as one.

Abstract

Lu et al. showed that ectopic expression of *Oct4*, *Sox2* and *Klf4* (OSK) in murine retinal ganglion cells restores youthful DNA methylation, promotes axon regeneration after optic nerve crush, and reverses vision loss, with the effect requiring the demethylases TET1 and TET2 [1]. This note asks what a model of the analogous intervention in *motor* neurons must contain, under the constraint that the target cells be hindbrain-derived, and it answers in the calculus of the companion notes: the rho calculus with graded where-clauses over $\mathbb{R}_{\geq 0}$ [14, 15], weighted GSLTs [16], the intervention membrane of [13], and the proofreading ladder of [17].

Four findings are reported. First, the published calibration does not transfer, and the formalism says why in a form that is checkable rather than rhetorical: a weight map is keyed by spatial–behavioural formulae refining a rule's redexes, so a change of *chromatin configuration* changes the key set and not merely the values, and the partition discipline of [16] renders an unclassified hindbrain redex a well-formedness violation rather than a poor fit. Retinal ganglion cells are forebrain-lineage; the bulbar motor pool is hindbrain-lineage; and Jokhai, Dundes et al. report that the anterior and posterior neural ectoderm arise as parallel progenitors with divergent chromatin landscapes [2]. A second transfer gap, from acute axotomy of a healthy neuron to chronic degeneration of a sick one, is larger and less tractable, and belongs in the term rather than in a coefficient.

Second, dynamic weight maps are necessary and not merely convenient. A locus with n CpG sites has 2^n methylation configurations; a static-rate stochastic π model must spend species on them, and SPiM is exactly the static-map, channel-identity-key restriction of the weighted construction [16]. Keying the map on a formula over the chromatin term prices the same state dependence at one species.

Third, rejuvenation and dedifferentiation are not two processes with two rates. They are candidates contending for the same OSK pool, hence members of one contention set, and the therapeutic window is therefore a property of the normalisation rather than of either rate — a first-passage property of the jump chain. Because the clause governs the jump chain and not the holding times, an induction schedule must be *reified as data* to come within reach of a do at all; a schedule left as an exogenous time-varying rate is outside the construct. Vector tropism is namespace confinement, and the schedule is a conditional σ-intervention in the dose reading of [13].

Fourth, the proofreading ladder yields a limitative result with an immediate experimental consequence. Pulsed induction is proofreading in time, its advantage over sustained

induction scales as a power of the single-stage discrimination factor f at the contested loci, and the exponent is independent of the number of pulses. Since energy buys confidence and never scope [17], no schedule manufactures selectivity that the OSK affinity landscape does not already supply. Measuring f on hPSC-derived hindbrain motor neurons is therefore a go/no-go test that precedes animal work and costs a binding assay rather than a cohort.

Four places where the apparatus is not ready are recorded rather than smoothed over: the pluripotency network's variable-level cycles, the permanent-shaped normalisation of a well-mixed cell's contention sets, structural identifiability, and the discipline required to distinguish a biologically real starvation from an over-restricted `do`. We close with what the model can and cannot license. Its job is to say which experiments to run and which protocol variants to discard — not which protocol to use.

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1 Introduction

1.1 The question

The optic-nerve result of [1] is the cleanest demonstration that partial reprogramming can restore regenerative competence to a central nervous system neuron without erasing its identity. Three factors suffice — Oct4, Sox2, Klf4, with c-Myc withheld on oncogenic grounds — delivered by AAV under a tetracycline-controlled switch, and the restoration of youthful methylation depends on TET1 and TET2.

The question this note addresses is what must be true of a *model* of the analogous intervention in motor neurons, under one anatomical constraint: that the cells be hindbrain-derived. The constraint is motivated by [2], which reports that the brain develops from two parallel progenitor populations rather than one, and that the posterior population — the hindbrain's — had resisted derivation in vitro precisely because it is not reachable from the anterior one.

The note does not propose an intervention. It specifies a model, locates it in a fragment of the calculus where its semantics is exact and its cost is bounded, derives from it the measurements it needs, and states what it is competent to conclude. The last of these is the load-bearing contribution: a model of a therapeutic cascade is worth having only in proportion to the sharpness with which its own limits are stated.

1.2 Why this calculus

Three properties of the graded where-clause programme make it the right host, and one of them is decisive.

The decisive one is that the central biological question is a question about *branches*. Partial reprogramming succeeds when a cell recovers regenerative competence and fails when it loses terminal identity, and these are competing outcomes of the same driving process. A model whose dynamics are confluent — a system of rate equations, a mean-field trajectory — can *state* that the second outcome has some probability, but it cannot *inhabit* the branching that makes the therapeutic window a window. This is the observation of [19] transposed: on a confluent witness language the linear-time/branching-time spectrum collapses, and the structure that a therapeutic window *is* lives in the part that collapsed.

The second is that the state dependence of the kinetics is a dependence on *shape*. Methylation state is a configuration of a locus, and the propensity of the next rewrite depends on it. A weight map keyed by spatial–behavioural formulae over the term expresses that directly (§4); a static rate attached to a channel identity cannot, and pays for the difference in species.

The third is that the intervention has a home. The construct $\mathsf{do}(C)\ \mathsf{in}\ P$ of [13] is a policy conditional on a race arising, scoped at the cut, which is the right shape for a therapeutic intervention: one does not know in advance which cell will be in which state when the vector arrives, so one installs a policy rather than performing a surgery on a named node.

1.3 Contributions

1. **A transfer obstruction with a checkable form** (§3). The anterior calibration does not carry to the posterior organ by reparameterisation, because the weight map's keys and not only its values change; the partition discipline makes the failure a well-formedness violation. A second obstruction — acute injury versus chronic degeneration — is identified as the larger of the two.

2. **Dynamic weight maps as a necessity** (§4). The 2^n argument, and the precise sense in which the model lies strictly beyond SPiM.

3. **The window as a contention-set property** (§5). Rejuvenation and dedifferentiation contend for one OSK pool; the window is a first-passage property of the normalised jump chain; and the schedule must be reified as data to be intervenable.

4. **A proofreading bound and a cheap test** (§6). Pulsing is proofreading in time; its benefit is a power of the single-stage discrimination factor; and the scope limitation of [17] converts into a go/no-go measurement that precedes animal work.

5. **Counterfactual cohort design** (§7). Replay under do with the exogenous record held gives matched-pair contrasts per cell, and the variance reduction is the practical payoff of rung three.

6. **Four gaps, and what they cost** (§8), and a statement of licensed and unlicensed conclusions (§9).

1.4 What this note does not do

It does not fit parameters; most of the relevant rate constants are not known and §8 argues that fitting before an identifiability analysis would produce a curve rather than a model. It does not treat vector biodistribution, capsid immunogenicity, or whole-organism outcome, all of which the model consumes as inputs. It does not discharge the cyclic case at variable level. It does not price a do against the ledger, which remains Obligation 5.3 of [13]. And it proposes no protocol.

2 Scope and substrate

2.1 The target compartment

The modelled substrate is the hindbrain-derived bulbar motor pool: the cranial motor nuclei — hypoglossal, facial, trigeminal motor, and the vagal/ambiguus column — arising from the *Gbx2*-expressing posterior neural ectoderm of [2]. The choice is not presentational. It fixes the weight maps, and §3 argues it blocks reuse of the published calibration.

2.2 Two populations, one delivery term

The constraint has a consequence that must be built in rather than noted. The medullary respiratory nuclei occupy the same compartment, so a model containing only the intended target population cannot represent the dominant risk at all: there is no term in it for the thing one is worried about.

Principle 2.1 (Off-target population is a modelled population)**.** The model carries at least two cell populations under one delivery term: the intended bulbar motor pool, and a neighbouring population receiving the same vector exposure and the same induction schedule but differing in

chromatin state and in the consequence of mis-reprogramming. A model with one population and a scalar "off-target rate" is not a model of the risk; it is a parameter named after it.

The relevance of the compartment is not incidental to the disease. [2] notes that amyotrophic lateral sclerosis affects both organs, so a hindbrain-restricted intervention is partial by construction, and the model should report what fraction of the motor deficit its target compartment can in principle address rather than leaving that fraction implicit.

2.3 Deliberate exclusions

Four things lie outside the model: in vivo vector biodistribution and pharmacokinetics; the immune response to the capsid; whole-organism functional outcome; and anything resembling a human dose. The first three are measurement problems the model consumes as inputs. The fourth is not a modelling question and does not become one by being asked of a model.

The modelled object is therefore one cell's cascade, instantiated across a population of two cell types, under a shared intervention term.

3 The transfer obstruction

3.1 Keys, not values

The retina arises from the optic vesicle, an evagination of the diencephalon, so retinal ganglion cells are forebrain-lineage neurons. The result of [1] therefore sits entirely within the anterior organ of [2], while the target of this programme sits in the posterior one.

In the weighted-GSLT setting this is not a change of parameters, and the reason is structural.

Observation 3.1 (A lineage change is a change of key set)**.** A weight map w assigns weights to keys that are spatial–behavioural formulae refining a rule's left-hand side [16]. The keys are predicates on the *shape* of the state. A reparameterisation changes $\mathsf{w}(\varphi)$ at fixed φ. A change of chromatin configuration changes which refinements are distinguishable — which φ there are — and so changes the domain of w, not its values.

The force of this is that the partition discipline makes the failure detectable rather than silent. The keys of a rule's map must be pairwise exclusive and jointly exhaustive over that rule's redexes; this single requirement is what makes the weighting a total, single-valued classification [16]. Hence:

Proposition 3.2 (Unclassified redexes are a well-formedness violation)**.** *Let w^A be a weight map whose partition was estimated on anterior substrate, and let P^H be a term over hindbrain substrate. If some redex of P^H satisfies no key of w^A, then w^A is not a weight map for P^H: the classification is not exhaustive, and no choice of values repairs it.*

So porting the anterior calibration and re-estimating rates has two possible outcomes, and the discipline is what distinguishes them: either the port throws, in which case the transfer obstruction is established, or it does not, in which case the anterior partition is exhaustive for the hindbrain term and the obstruction is confined to the values. Either answer is informative, and the test is cheap because it is static.

3.2 The larger gap: acute injury versus chronic disease

The lineage gap is the one the formalism makes crisp. The injury-regime gap is probably the one that decides whether the approach works, and it has had less attention.

The published result is axon regeneration after acute optic nerve crush in an otherwise healthy neuron, together with restored function in aged and glaucomatous eyes. Amyotrophic lateral sclerosis is chronic, progressive degeneration of a cell under sustained proteostatic and excitotoxic stress, with the neuromuscular junction failing distally. "Reprogramme a healthy cell whose axon was severed" and "reprogramme a sick cell that is dying slowly" are different computational problems.

Principle 3.3 (The disease belongs in the term)**.** A model that represents the diseased baseline as a reduced regeneration rate will overstate the intervention's reach, because it deletes the competition. The pathological processes are contending candidates at the same sites the intervention acts on — they consume the same proteostatic capacity and the same transcriptional machinery — so they belong in the parallel composition, not in a coefficient.

This is the point at which the branching argument of §1 does real work rather than decorative work. In a rate-equation model the disease and the therapy are additive terms on one trajectory. In the term model they are candidates in contention, and the quantity of interest is which fires.

3.3 The empirical assumption this rests on

The argument of §3.1 rests on one claim the source does not establish: that the divergent chromatin landscape of the two progenitor populations *persists into the mature motor neuron.* [2] is a result about progenitors at gastrulation. If the divergence largely washes out by maturity, the lineage objection weakens substantially and the anterior calibration becomes a defensible prior.

The two-organ claim is also recent and contested at its edges; the distinctness of the two progenitor populations has been questioned in commentary on the paper. The model should therefore carry lineage divergence as a quantity with a magnitude and an uncertainty rather than as a binary, and that magnitude is among the first things worth measuring, since the whole question of what may be reused turns on it.

Obligation 3.4 (Persistence of divergence)**.** Determine whether the chromatin divergence reported between anterior and posterior neural ectoderm persists in terminally differentiated motor neurons, and at which loci. The deliverable is not a yes/no but a per-locus magnitude, because the weight map's keys are per-locus predicates.

4 The model, in four layers

The model is built in four layers, each of which is a known fragment of the programme. The layering is not stylistic: each layer has a different cost profile and a different fragment discipline, and conflating them is how such models become intractable.

4.1 Layer 0: species, reactions, propensity

The base layer is ordinary stochastic chemistry: AAV particles, tTA and TRE-OSK transcripts, OSK protein, TET1/TET2, the methylation state of the modelled loci, and the pathological species of Principle 3.3. Receipts are persistent for-comprehensions, propensity is $\mathsf{a} = \mathsf{w}(\varphi) \cdot \sum_k \gamma(k)$ over classified, funded redex positions, and the simulator is Gillespie [6, 16].

Remark 4.1 (SPiM is the baseline, and it is free). Restricted to the π-calculus with static maps and channel-identity keys, the weighted construction is *exactly* the Stochastic Pi Machine [16, 4]. Layer 0 is therefore a SPiM model in the precise sense, and that is a deliberate choice: it is the layer that collaborators outside this programme can read, check and run in their own

tooling, and it costs nothing to expose because the theorem already identifies it as a restriction of what is being built.

4.2 Layer 1: dynamic weight maps for epigenetic state

Layer 0 cannot express the dependence that matters. The propensity of demethylation at a locus depends on the current methylation configuration of that locus and of its neighbours — cooperativity, spreading, CpG-island boundary effects. In a static-rate model this dependence must be spent in species.

Observation 4.2 (The 2^n argument)**.** A locus with n CpG sites has 2^n methylation configurations. A model whose rates are attached to channel identities and are static must introduce one species (or one equivalence class of species, chosen in advance) per configuration on which a rate depends. For a promoter region with twenty CpG sites this is of order 10^6 species. A weight map keyed on a spatial–behavioural formula over the chromatin term — "at least k of the sites in this window are unmethylated", "this boundary element is unoccupied" — expresses the same dependence with one species, because the configuration is in the term and the key is a predicate on it.

This is where "dynamic maps take the construction strictly beyond SPiM" [16] stops being a slogan about expressiveness and becomes the reason the model is writable at all. Weight-map instances are part of the state specification, so a rewrite may update them: methylation changes the map that governs subsequent methylation, which is what an epigenetic model is.

Obligation 4.3 (The key set is the modelling work)**.** Choosing the refinement formulae — which shape predicates the model distinguishes — is the substantive modelling decision, and it is where domain knowledge enters. The partition discipline constrains it (exclusive, exhaustive) but does not determine it. An over-fine partition is unidentifiable (§8); an over-coarse one deletes the mechanism.

4.3 Layer 2: the intervention

The intervention has three parts, and each lands in a different place in [13].

Tropism is namespace confinement. The vector reaches the cells it reaches. In the calculus this is discipline $\mathcal{T}_4$: every constrained receipt lies within the deployer's seal, so the `do` reaches only the code its author paid for. The capability to intervene is the capability to deliver, and Proposition 2.3 of [13] — that no context hard-intervenes over $\mathbb{R}_{\geq 0}$ — is the formal counterpart of the fact that an untransduced cell is not reprogrammed.

The schedule is a conditional σ-intervention. The induction regime is not a hard `do`. It is a graded, conditional, scoped restriction: *if* a race for OSK occupancy arises in a transduced cell, resolve it with weights set by the current state of the switch. In the taxonomy of [13] this is a conditional σ-intervention with constraint in $\mathcal{F}_1$, and its induced parent set is read off the free names of the constraint — which is to say, the schedule's dependencies are written in the schedule.

```
do( occ |= dose(*clock) ) in { cell_1 | ... | cell_m | clock }
```

layer	what it carries	fragment	cost
L0	species, reactions, propensity	static maps, channel keys (= SPiM)	Gillespie step
L1	epigenetic state dependence	dynamic maps, $\mathcal{F}_1$ keys	$O(2^d)$ per candidate, d = formula depth
L2	tropism, schedule, dose	$\mathcal{T}_4$, $\mathcal{F}_1$, schedule reified	one COMM plus a conjunct
L3	locus discrimination	base rules, funded	tokens per discard step

Table 1: The layers, their fragments, and what each costs. Fragment and discipline names are those of [13].

The schedule must be reified as data. This is the non-obvious requirement, and it is forced.

Proposition 4.4 (An exogenous schedule is outside the construct)**.** *The clause governs the jump chain and not the holding times [15, Rem. 3.2], so an intervention on* when *a site fires is not expressible as a clause restriction. An induction schedule is precisely an intervention on when. Hence a schedule supplied as an exogenous time-varying rate lies outside the reach of any* do*, and the do-calculus results of [13] do not apply to it. Reifying the schedule as a process emitting tokens on a clock channel returns it to the clause's reach, because the consumption of a clock token is itself a race.*

Proof. Immediate from the cited remark together with Definition 3.2 of [13]: the membrane reads $\psi_r \otimes \lceil\varphi\rceil$ at receipts binding constrained variables, and a holding time is not a candidate at such a receipt. Once the schedule is a message on a channel, the receipt that consumes it binds a variable, and the variable is constrainable. □

The practical content of Proposition 4.4 is that schedules become *terms*, so comparing two induction regimes is comparing two terms under the same model, and the comparison is an interventional contrast rather than two separate simulations with different parameter files. That is also what makes the identifiability question of §8 well posed.

4.4 Layer 3: fidelity

The fourth layer is the discrimination between intended and unintended OSK targets, treated as a kinetic proofreading problem. It is developed in §6, because its content is a limitative result rather than a modelling convenience.

4.5 Where each layer runs

Table 1 records the architectural decision that §8 will defend: the do membrane is confined to Layer 2's designated sites, and the bulk chemistry of Layer 0 runs underneath it without one. Putting the whole cell under a do is the mistake that makes the model intractable.

5 The central race

5.1 Rejuvenation and dedifferentiation contend

Let $\mathcal{R}_{\text{rej}}$ be the rule class that demethylates regeneration-competence loci, restoring the youthful pattern, and $\mathcal{R}_{\text{ded}}$ the class that activates the pluripotency network and costs the cell its terminal identity. Both are driven by the same OSK pool.

Observation 5.1 (One contention set, not two rates)**.** Two candidates contend when they share a message or share a linear receipt [13, Def. 2.4]. A candidate in $\mathcal{R}_{\text{rej}}$ and a candidate in $\mathcal{R}_{\text{ded}}$ that consume the same OSK molecule share a message, hence lie in one connected component of the contention relation, hence in one contention set — which is the unit at which weights are normalised.

The consequence is the structural claim of this note.

Proposition 5.2 (The window is a normalisation property)**.** *Under Observation 5.1 the probability that a cell reaches the rejuvenation threshold before the dedifferentiation threshold is a first-passage property of the jump chain whose transition probabilities are the contention-set-normalised clause values, and it is not a function of the two rule classes' absolute propensities separately. Scaling both by a common factor leaves it unchanged; scaling either alone changes it.*

Proof. Resolution normalises within a contention set [13, Def. 2.6]: the probability that a maximal matching fires is its weight divided by the total over the set. A common positive scaling of all weights in a set leaves every ratio fixed, hence leaves the jump chain fixed; first passage is a property of the jump chain alone, the holding times being irrelevant to which threshold is reached first. □

Proposition 5.2 is worth stating because it says which experimental quantity matters and which does not. Total OSK expression level, raised or lowered uniformly, does not move the window; it moves the time scale. What moves the window is the *ratio* of propensities at the contested sites. An intervention that increases OSK abundance while leaving that ratio alone buys speed and no safety, which is a prediction that contradicts the intuition that lower dose is safer.

Remark 5.3 (Where the intuition is right anyway). Lower total exposure does reduce risk, but by a different route: it reduces the number of firings, hence the number of opportunities for a low-probability $\mathcal{R}_{\text{ded}}$ event, and it shortens the interval during which identity is destabilised. That is a statement about the *number of trials*, not about the per-trial odds, and the model should keep the two separate because they respond to different design levers.

5.2 What the cell's threshold is

Both thresholds require a definition in the term, and neither is obvious. The rejuvenation threshold is a predicate on the methylation configuration of the regeneration-competence loci — a formula in the crisp sort, of the kind Layer 1's keys already use. The dedifferentiation threshold is harder, because loss of identity is not a single locus event; it is plausibly a fixed-point property of the pluripotency network's mutual activation, which lands it in $\mathcal{F}_3$ and hence, by Principle 4.6 of [13], in the model checker rather than in the clause.

Obligation 5.4 (Thresholds as formulae)**.** Give both thresholds as formulae in the generated logic over the chromatin term, with the dedifferentiation threshold stated in $\mathcal{F}_3$ and lowered to $\mathcal{F}_1$ on the finite-state fragment by unfolding, per Definition 4.4 of [13]. Until this is done, the simulator has no stopping condition and the window has no definition.

6 Fidelity: a proofreading bound and a cheap test

6.1 Pulsing is proofreading in time

OSK must bind the right loci. The discrimination is between binding sites that differ by a small free-energy margin, which is the classical setting of Hopfield–Ninio kinetic proofreading.

The companion note establishes the relevant structure: a catalytic step is a context rule, restoring its left-hand side and contributing a multiplicative factor at no primitive cost, while a proofreading step is a base rule, consuming its redex and clearing the funding judgement [17]. Selectivity beyond the binding energies requires the second kind.

A pulsed induction regime has exactly that shape. Each withdrawal phase releases complexes that have not committed, which is a discard step; each subsequent induction phase re-draws. The ladder's rungs are pulses and its currency is expression.

Observation 6.1 (The exchange rate)**.** By the exchange rate of [17], error falls as a power of the resource burned, with an exponent determined by single-stage kinetics and independent of the depth of the ladder. Writing f for the single-stage discrimination factor at the contested loci — the ratio of off-rates for incorrect versus correct occupancy — the achievable error after n discard stages scales as f^{-n} in the ideal limit, with the realised exponent degraded by re-binding during the off phase.

Prediction 6.2 (Pulsed versus sustained)**.** Pulsed induction improves selectivity over sustained induction if and only if $f > 1$ at the contested loci, and the magnitude of the improvement is governed by f and by the ratio of the off-phase duration to the re-binding time, not by the number of pulses beyond the point at which re-binding dominates. If $f \approx 1$, schedule optimisation is not the design problem.

Prediction 6.2 is ordinal and falsifiable, and it does not require any of the unknown rate constants of §8: it requires f, which is a two-parameter binding measurement.

6.2 The limitative result, and what it implies for the programme

The sharper content is negative.

Principle 6.3 (Energy buys confidence and never scope)**.** A proofreading step is a move in the fibre [17]. It can increase the reliability with which a discrimination already available is made; it cannot create a discrimination that the affinity landscape does not support. Consequently no induction schedule, however elaborate, manufactures selectivity between regeneration-competence loci and pluripotency loci if OSK does not distinguish them.

The programmatic consequence is a fork, and it is worth facing before any animal work.

1. If f is comfortably greater than one at the contested loci, schedule design is the right lever, Prediction 6.2 tells you which direction to push, and the model's job is to find the schedule that maximises the window subject to the trial-count consideration of Remark 5.3.
2. If f is near one, no schedule helps, and the design problem is a different one: a different factor set, or a targeting layer that supplies the discrimination exogenously — locus-directed recruitment rather than diffuse overexpression. The model would then be a model of the wrong intervention, and the right response is to change the intervention rather than to tune it.

Principle 6.4 (The go/no-go precedes the model)**.** Measuring f on hPSC-derived hindbrain motor neurons [2] is a binding assay on the correct substrate. It is cheaper than a cohort, it is upstream of every parameter the model lacks, and by Principle 6.3 it determines which of the two programmes above is the one being pursued. It should therefore be the first bench work, before model fitting and before animal work.

This is the note's most actionable finding, and it is a consequence of the formalism rather than of the biology: the limitative theorem is what tells you that this particular measurement screens off the rest.

7 Counterfactual cohort design

Rung three of the ladder has a concrete payoff here, and it is statistical rather than philosophical.

A run of the model is a causal history together with the candidate that fired at each site; its exogenous record is the vector of resolver draws, and a run is replayed under $\mathsf{do}(C)$ by holding that record and re-resolving only where the restriction bites [13, Def. 3.12, Thm. 3.13]. For a simulated cell this answers: *would this cell have degenerated without the intervention?* — with the same noise realisation on both arms.

Observation 7.1 (Matched pairs for free)**.** Replay gives a per-cell matched-pair contrast rather than a difference of cohort means. The variance of the estimated treatment effect is accordingly the variance of the within-pair difference, which removes the between-cell variation that dominates a two-cohort comparison of heterogeneous neurons.

Two cautions. First, the clean statement of Theorem 3.13 of [13] holds on the staged fragment $\mathcal{T}_1$, where every gadget's contention set is itself and the recorded class is always still a candidate; a well-mixed cell is not staged, so off that fragment the "and otherwise sample" branch is taken and the matching degrades. Quantifying how much it degrades, for this model, is Obligation 7.2. Second, the variance reduction is a property of the *simulation*, not of a real cohort: no animal study has access to the exogenous record. The payoff is in designing the study — power calculations, which readouts at which timepoints — not in analysing it.

Obligation 7.2 (Replay fidelity off the staged fragment)**.** Measure, in simulation, the fraction of replayed firings at which the recorded candidate is no longer available, as a function of the intervention's position in the cascade. This bounds the usefulness of counterfactual contrasts for this model and is cheap to obtain once Layer 0 runs.

8 Four places the apparatus is not ready

Each of these is a real gap. Each has a workaround with a stated cost. None is fatal to the programme, and two of them bound what the model may claim.

8.1 Cycles at variable level

The pluripotency network is mutually activating: Oct4, Sox2 and Nanog reinforce one another, and the autoregulation is the mechanism by which dedifferentiation, once begun, completes. So the variable-level graph has cycles, while Pearl's apparatus and the surgery theorem of [13] are stated where the graph is a DAG.

The distinction the companion note draws is the one to use: the *event-level* graph is always acyclic, because an event structure's configurations are partial orders; the *variable-level* graph, which identifies events of a kind, may cycle when the term recurses. The workaround is to work at event level with unrolling depth equal to the inference budget β, treating the variable-level cycle as its unrolling.

The cost is that every interventional claim is indexed by β, and the β-dependence must be reported rather than suppressed. A claim that stabilises in β is trustworthy; one that does not is an artefact of the truncation. Obligation 5.6 of [13] — cyclic SCMs meeting the graded fixed point [11] — is the principled route and is open.

8.2 Well-mixed contention sets

A cell is a beaker. There are many copies of each species, so contention sets are large, and the normaliser is a sum over maximal matchings, which is permanent-shaped and #P-hard in general [13, §4.1]. Discipline $\mathcal{T}_1$ buys singleton contention sets, but a cell does not satisfy it.

The architecture of Table 1 is the workaround. The Gillespie propensity reading handles multiplicity directly, as the $k = 1$ truncation — the ideal gas of [18] — where propensity is a count times a rate and no permanent arises. The do membrane with $\mathcal{F}_0$ or $\mathcal{F}_1$ constraints is confined to Layer 2's designated intervention sites, which are few and whose contention sets are small by construction.

Principle 8.1 (Do not put the cell under a do)**.** The membrane is scoped to the sites the intervention acts on. Bulk chemistry runs beneath it. A do whose scope is the whole cell forces recomputation of large contention-set normalisers and is the proximate cause of intractability in models of this shape.

Spatial structure — nuclear compartmentalisation, condensate formation — would require the collision kernel of [18] and its higher virial terms. That machinery should be deferred: it is not needed for the questions of §5 and §6, and importing it early would cost the identifiability of §8.3 for no gain.

8.3 Identifiability, which is the real risk

Most of the rate constants are unknown, and this is where the project fails if it fails. The gradient machinery is available: over $\mathbb{R}_{\geq 0}$ a modality-free clause is a posynomial in its parameters, hence smooth and log-convex in log-coordinates, the likelihood factors over firing sites, and the parameter gradient decomposes into local scores computable at each site from candidate values the site already had [15]. Fitting is therefore mechanically straightforward.

Fitting is not identification.

Principle 8.2 (Identifiability analysis precedes estimation)**.** A model with many unidentifiable parameters that reproduces the available data is a curve, not a model: it will fit, it will interpolate, and its interventional predictions — the only predictions wanted here — will be determined by the prior rather than the data. Structural identifiability analysis therefore precedes parameter estimation, and its output is a ranked list of candidate measurements by information gain about the quantities §5 and §6 actually use.

The sharp form of the question is Obligation 5.4 of [13]: Pearl's identifiability problem rendered as a question about which interventional quantities are recoverable from the observational data one can actually obtain. Here the observables are realistic and limited — single-cell transcriptomes at a few timepoints, methylation arrays, immunostaining, occupancy assays — and the hidden variables are the states no assay reports. Transferring the ID algorithm [10] to the staged fragment is the stated route and is undischarged.

Obligation 8.3 (Identifiability before fitting)**.** Perform structural identifiability analysis on the Layer 0 and Layer 1 model before estimating any parameter, and report which of Proposition 5.2's ratio and Observation 6.1's exponent are identifiable from which observable sets. This is the deliverable on which the model's usefulness depends.

8.4 Starvation discipline

A restricted receipt with no satisfying candidate waits forever: the history deadlocks by starvation, which is how the embedding renders the failure of Pearl's positivity condition [13, Rem. 3.8]. In a biological model a starvation may be a real dead end — a cell state from which the cascade cannot proceed — or an artefact of an over-restricted do. These must not be confused, and in simulation they look identical.

Under discipline $\mathcal{T}_3$, where every channel a constrained variable binds on carries a type with finitely many closed values, vacuity and fatality are decidable by inspecting which values the clause weights positively [13, Def. 4.5]. The discipline to adopt is accordingly:

Principle 8.4 (Every do ships with its vacuity check)**.** Each intervention term in the model carries a decidable vacuity-and-fatality check, discharged statically under $\mathcal{T}_3$. Any starvation observed in simulation is triaged against that check before being interpreted biologically.

9 What the model can and cannot license

This section is the one to agree before bench work, because it is the boundary that makes the rest usable.

9.1 Licensed

- **Ranking schedules.** Ordinal comparison of induction regimes by the window of Proposition 5.2, under stated kinetics, with the sensitivity of the ranking reported alongside it.
- **Prioritising measurements.** Identification of which parameters the ranking depends on, hence which assays buy the most information per unit cost (Obligation 8.3).
- **Falsifiable ordinal predictions.** Prediction 6.2 is testable without knowing the unknown rates.
- **Discarding variants.** A protocol variant that the model shows to be dominated under every parameterisation consistent with the data need not be tested. Negative screening is where a model of this kind earns its keep.
- **Detecting structural problems early.** Non-identifiability and transfer obstructions (Proposition 3.2) are found statically, before money is spent.
- **Study design.** Counterfactual contrasts for power calculation and readout timing (§7).

9.2 Not licensed

- **Safety.** The model cannot establish it. The events that matter — transformation, loss of identity in a critical nucleus — are rare, their governing parameters are the least known in the model, and rare-event probabilities are precisely where a model with unidentified parameters is least trustworthy. A simulated window is not a safety margin.
- **Magnitudes of oncogenic risk.** Withholding c-Myc reduces a risk that the model cannot quantify, because quantifying it requires the tail of a distribution the model does not constrain.
- **Substituting for toxicology or animal work.** The model reorders and prunes that work; it does not replace any of it.
- **Any dose, route, or schedule for administration to a person.** Out of scope by §2.3 and not recoverable from anything here.

Principle 9.1 (What the model is for)**.** The model's job is to say which experiments to run and which protocol variants to discard. It is not to say which protocol to use.

10 A staged programme

The ordering below is chosen so that each stage produces something checkable and so that the stages most likely to invalidate the approach come early.

S0 Substrate and populations. Fix the bulbar motor pool and the off-target population of Principle 2.1; enumerate species and loci; state both thresholds (Obligation 5.4).

S1 Layer 0, as a SPiM model. Static maps, channel-identity keys. Literature rates where they exist, explicit `unknown` flags where they do not. Readable and runnable outside this programme by Remark 4.1.

S2 Layer 1, the dynamic-map lift. The first publishable unit; see below.

S3 Layer 2, the intervention. Schedule reified per Proposition 4.4; tropism as $\mathcal{T}_4$; schedules as terms.

S4 Layer 3 and the f measurement. The go/no-go of Principle 6.4. *This is the first bench work.*

S5 Identifiability and sensitivity. Obligation 8.3. The deliverable for experimental collaborators.

S6 Cohort design. §7, conditional on Obligation 7.2.

Two remarks on the ordering. The f measurement (S4) is placed before identifiability and fitting because Principle 6.3 says it screens off which programme is being pursued; spending on model calibration before knowing the answer risks calibrating a model of the wrong intervention. And S1 is retained as a deliverable in its own right rather than folded into S2, because an externally checkable artefact is worth more than the week it costs.

10.1 The first publishable unit

S2 is the natural first paper: *dynamic weight maps for epigenetic state, with a reprogramming cascade as the worked example.* It is a formal result about expressiveness — Observation 4.2, with the SPiM restriction theorem as its baseline — and so does not depend on the unknown rate constants. It is also the piece that makes the transfer obstruction of §3 statable, since that argument is about key sets.

The transferability argument itself is a plausible second, and a shorter one: *why a model calibrated on anterior-lineage CNS neurons does not transfer to posterior-lineage motor neurons.* It would be read by a different audience than this programme's usual one, it makes a prediction the hPSC substrate of [2] can test directly, and Proposition 3.2 gives it a sharper form than the biological literature currently has.

11 Literature points to settle first

Four factual matters bear on the model's construction and should be confirmed against the primary sources rather than carried from summaries.

1. **Switch polarity.** The system of [1] uses tTA under a constitutive promoter with TRE-OSK, and the primary material describes induction in the *absence* of doxycycline, which is Tet-Off. Secondary summaries state the opposite. The polarity determines the sign of the schedule term in Layer 2 and must be taken from the primary source.
2. **TET1/TET2 dependence across lineages.** The requirement for the demethylases is established on anterior substrate. Whether it holds, and with what stoichiometry, in posterior-lineage motor neurons is a parameter with a lineage index, and by Observation 3.1 possibly a key with one.

3. **Which motor pools are in scope.** Spinal motor neurons derive from neuromesodermal progenitors rather than from hindbrain neural ectoderm, so the hindbrain constraint restricts the target to the cranial motor nuclei. If the intent is to include spinal pools, that is a third lineage and a third partition, not an extension of the second.

4. **Current clinical status of the anterior programme.** An OSK-based therapy derived from this work has entered human trials for glaucoma. What has been published about its dosing and safety profile is directly relevant to Layer 2 and should be tracked, with the caveat that it is an anterior-organ, acute-accessible-tissue indication.

12 Conclusion

The apparatus fits this problem better than it had any particular right to, and the fit is not coincidental: a therapeutic window is a statement about which of two competing futures a cell enters, and that is the structure the graded where-clause was built to make writable. Rejuvenation and dedifferentiation contend for one OSK pool, so the window is a property of the contention set's normalisation rather than of either propensity alone, and raising expression uniformly buys time rather than safety. The induction schedule must be reified as data before any intervention operator can reach it. Epigenetic state dependence needs dynamic weight maps, which is where the construction leaves SPiM behind, and SPiM remains available underneath as the layer collaborators can check.

Two results point outward rather than inward. The published calibration does not transfer across the organ boundary, and the partition discipline makes that a well-formedness question rather than a matter of opinion — with the persistence of progenitor divergence into maturity as the empirical assumption the argument rests on. And the proofreading ladder, read for its limitative content, says that no schedule creates selectivity the affinity landscape does not supply, which converts a theorem into the cheapest consequential experiment available: measure the single-stage discrimination factor on hindbrain motor neurons, and let it decide which of two different programmes this is.

The model will not establish that anything is safe. What it can do is spend the next year's experiments in a better order than intuition would, and say plainly which of its own conclusions are load-bearing and which are furniture. That is the whole of what is on offer, and it is worth having.

References

[1] Y. Lu, B. Brommer, X. Tian, A. Krishnan, M. Meer, C. Wang, D. L. Vera, Q. Zeng, D. Yu, M. S. Bonkowski, J.-H. Yang, S. Zhou, E. M. Hoffmann, M. M. Karg, M. B. Schultz, A. E. Kane, N. Davidsohn, E. Korobkina, K. Chwalek, L. A. Rajman, G. M. Church, K. Hochedlinger, V. N. Gladyshev, S. Horvath, M. E. Levine, M. S. Gregory-Ksander, B. R. Ksander, Z. He, and D. A. Sinclair. Reprogramming to recover youthful epigenetic information and restore vision. *Nature* 588:124–129, 2020. https://doi.org/10.1038/s41586-020-2975-4

[2] R. Jokhai, L. Dundes, et al. (K. M. Loh, senior author). Two parallel lineage-committed progenitors contribute to the developing brain. *Nature Neuroscience*, published 18 September 2026. Preprint: *bioRxiv* 2025.07.02.662771. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12236623/

[3] A. Ocampo, P. Reddy, P. Martinez-Redondo, A. Platero-Luengo, F. Hatanaka, T. Hishida, et al. In vivo amelioration of age-associated hallmarks by partial reprogramming. *Cell* 167(7):1719–1733, 2016.

[4] A. Phillips and L. Cardelli. Efficient, correct simulation of biological processes in the stochastic pi-calculus. In *Computational Methods in Systems Biology*, LNCS 4695, pp. 184–199, 2007.

[5] A. Phillips and L. Cardelli. A correct abstract machine for the stochastic pi-calculus. In *Concurrent Models in Molecular Biology (BioCONCUR)*, 2004.

[6] D. T. Gillespie. Exact stochastic simulation of coupled chemical reactions. *Journal of Physical Chemistry* 81(25):2340–2361, 1977.

[7] J. J. Hopfield. Kinetic proofreading: a new mechanism for reducing errors in biosynthetic processes requiring high specificity. *PNAS* 71(10):4135–4139, 1974.

[8] J. Ninio. Kinetic amplification of enzyme discrimination. *Biochimie* 57(5):587–595, 1975.

[9] J. Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, second edition, 2009.

[10] I. Shpitser and J. Pearl. Identification of joint interventional distributions in recursive semi-Markovian causal models. In *AAAI 2006*, pp. 1219–1226.

[11] S. Bongers, P. Forré, J. Peters, and J. M. Mooij. Foundations of structural causal models with cycles and latent variables. *Annals of Statistics* 49(5):2885–2915, 2021.

[12] L. G. Meredith and M. Radestock. A reflective higher-order calculus. *Electronic Notes in Theoretical Computer Science* 141(5):49–67, 2005.

[13] L. G. Meredith. Interventions at the cut: Pearl's `do` as a scoped guard in the rho calculus with graded where-clauses. F1R3FLY.io working note, `fuzzyware/interventions-at-the-cut`, November 2026.

[14] L. G. Meredith. Graded where-clauses: non-crisp truth values as the syntax of non-determinism resolution in the rho calculus, version 4. F1R3FLY.io working note, `fuzzyware/graded-where-v4`, August 2026.

[15] L. G. Meredith. Graded where-clauses over the reals: one-dimensional grading as the mortal scientist's gradient, and probabilistic logic as the metric that gradient requires. F1R3FLY.io working note, `fuzzyware/graded-where-pln`, August 2026.

[16] L. G. Meredith. Weighted GSLTs: stochastic and quantum simulation for decorated continued interactive GSLTs. F1R3FLY.io working note, `rho-life/weighted-gslt`, 2026.

[17] L. G. Meredith. Kinetic proofreading in the mortal scientist setting: energy into accuracy, and the price of commitment. F1R3FLY.io working note, `kinetic-proofreading/proofreading-note`, August 2026.

[18] L. G. Meredith. Process terms as molecules: locations, collisions, and the road to viscosity. F1R3FLY.io working note, `fuzzyware/process-molecules`, 2026.

[19] L. G. Meredith. Inhabiting choice: what a confluent witness cannot be. F1R3FLY.io working note, `proof-dynamics-and-reasoning/inhabiting-choice`, November 2026.

[20] L. G. Meredith. Proof search as concurrent reaction. F1R3FLY.io working note, `proof-dynamics-and-reasoning/proof-search`, 2026.

[21] L. G. Meredith. Cost accounting as an endofunctor on continued interactive GSLTs. F1R3FLY.io working note, `cost-accounting-as-monad/continued-gslt-cost-v2`, 2026.